



Strength of Materials II



Chapter Two: Curved Beam

MEng3111



Curved Beam

- Thus far, we have only analyzed stress in straight beams.
- However, there are many situations where curved beams are used.



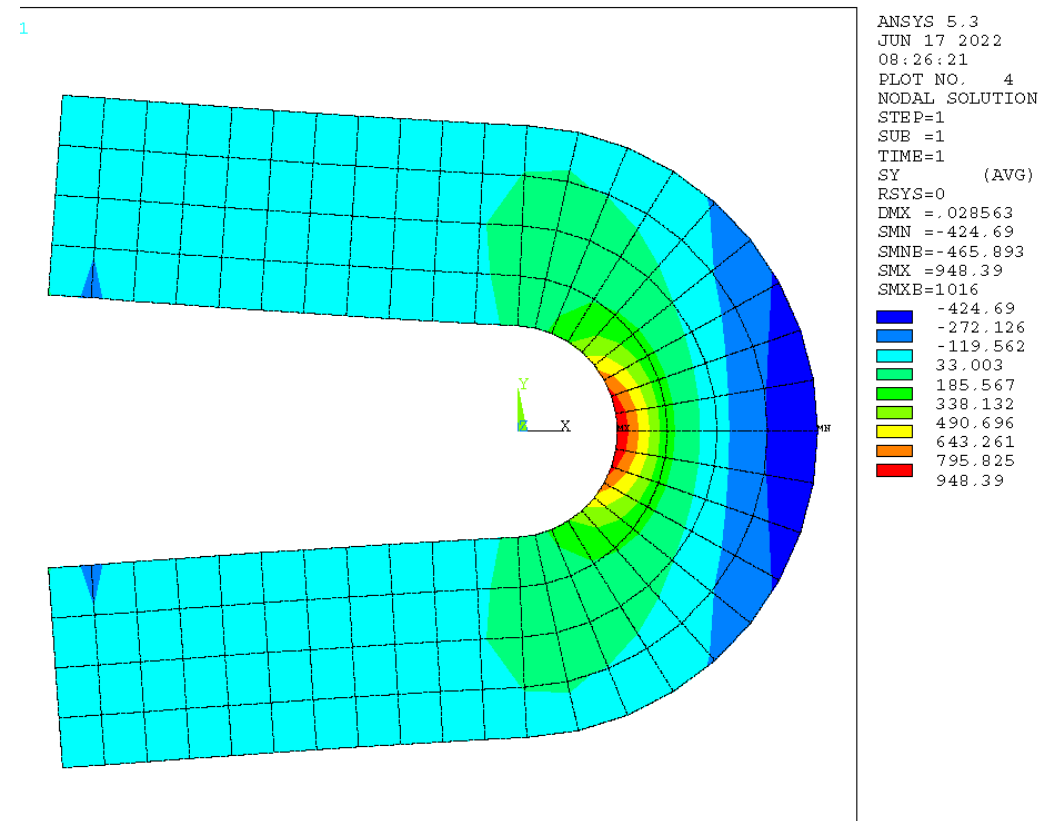
Curved structural beams

Hooks



Chain links

This chapter on Curved Beams in Bending introduces the necessary theory and formulas for analyzing components with significant curvature, a scenario where the simpler linear analysis for straight beams is inadequate.



Objective

- Determine the bending stresses due to the application of equal and opposite couples to a member that is initially curved.
- Discussion is limited to curved members with uniform cross sections possessing a plane of symmetry in which the bending couples are applied.
- It is assumed that all stresses remain below the proportional limit.

Curved Beam

Assumptions:

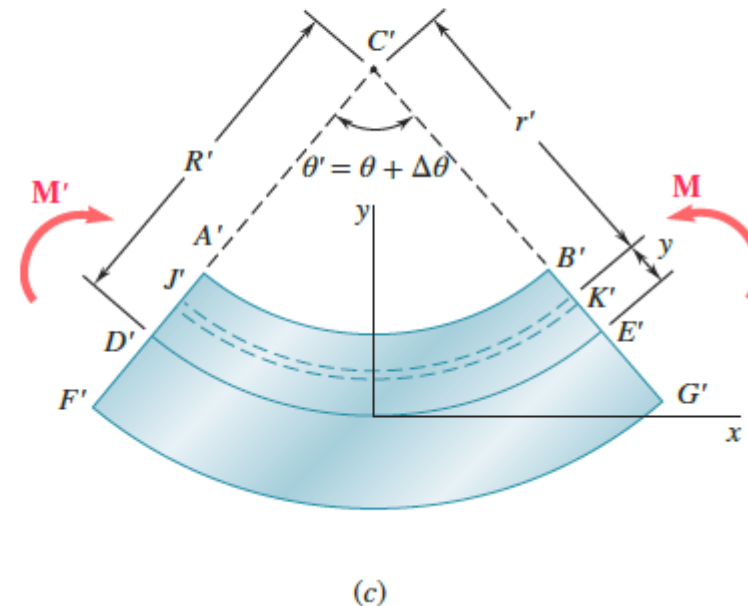
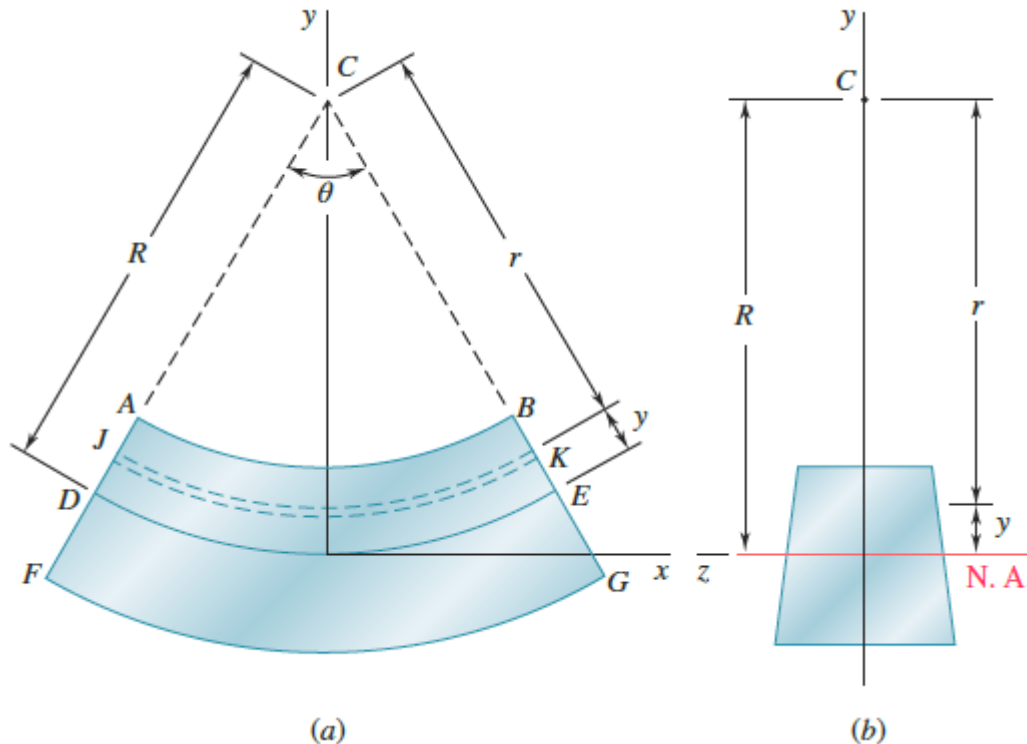
- Pure bending (no shear and axial forces present - will add these later)
- Bending occurs in a single plane
- The cross-section has at least one axis of symmetry
- Material is homogeneous and isotropic and behaves in linear-elastic manner under loading

What does this mean?

- $\sigma = -My/I$ no longer applies
- Neutral axis and axis of symmetry (centroid) are no longer the same
- Stress distribution is not linear

Curved Beam

- Consider the curved beam of uniform crosssection (Symmetric with respect to y axis)
- In its unstressed state, its upper and lower surfaces intersects the vertical XY plane along arcs of the circle AB and FG centered at C
- Opposite couples M and M' in the plane of symmetry

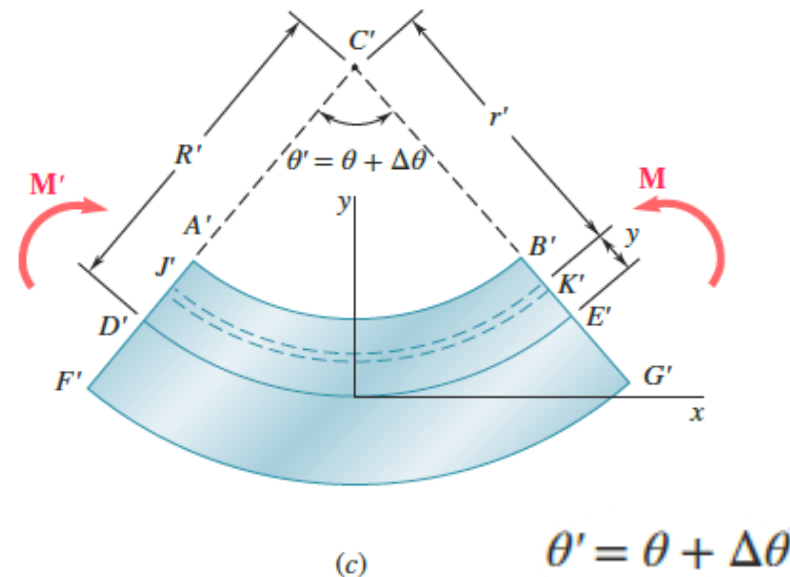
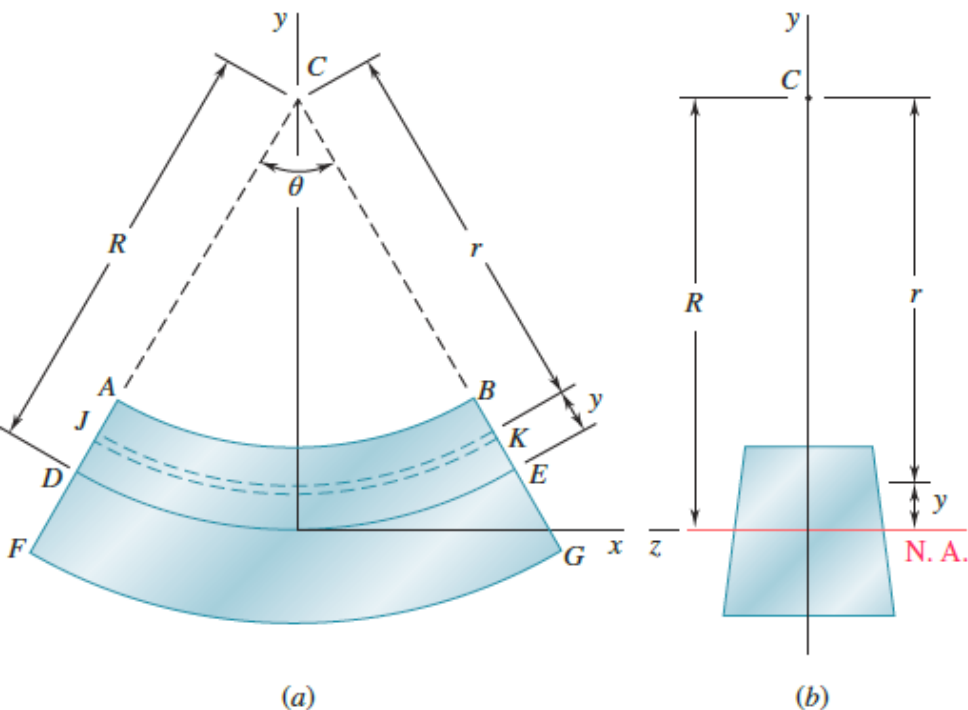


Key Parameters

Parameter	Symbol	Description
Centroidal Radius	$\bar{r}$ (or r_c)	Distance from the center of curvature (C) to the Centroid of the cross-section.
Neutral Axis Radius	R	Distance from the center of curvature (C) to the Neutral Surface .
Eccentricity	e	The distance between the centroid and the neutral axis: $e = \bar{r} - R$.

Curved Beam

- $A'C' < AC$
- Also, the couples M and M' cause the length of the upper surface of the member to decrease ($A'B' < AB$)
- length of the lower surface to increase ($F'G' > FG$). Therefore, we conclude that a **neutral surface** must exist in the member, the length of which remains constant.



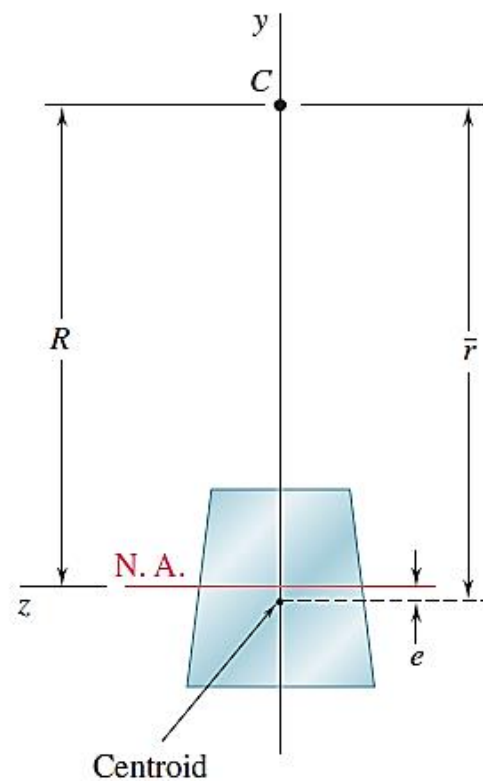
$$R\theta = R'\theta'$$

Distance R from the center of curvature C to the neutral surface

$$R = \frac{A}{\int \frac{dA}{r}}$$

Curved Beam

Distance R from the center of curvature C to the neutral surface



$$R = \frac{A}{\int \frac{dA}{r}}$$

$$\bar{r} = \frac{1}{A} \int r dA$$

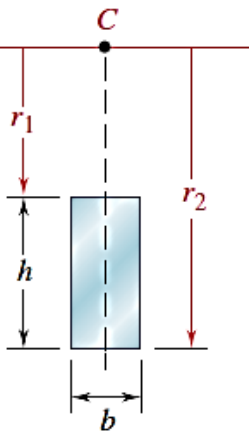
$$\bar{r} - R = e$$

$$\frac{E \Delta \theta}{\theta} = \frac{M}{A(\bar{r} - R)}$$

A : Area of the cross-section.

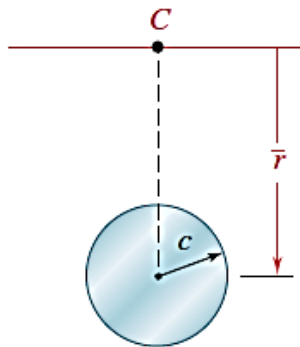
dA : Elemental area.

r : Radial distance from the center of curvature to the elemental area dA .



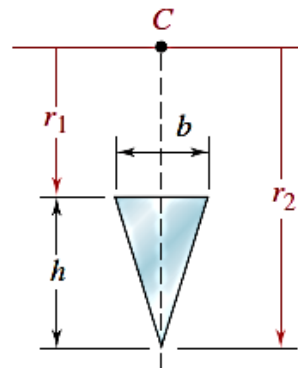
Rectangle

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$



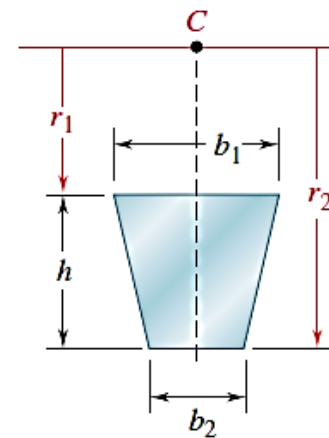
Circle

$$R = \frac{1}{2} (\bar{r} + \sqrt{\bar{r}^2 - c^2})$$



Triangle

$$R = \frac{\frac{1}{2}h}{\frac{r_2}{h} \ln \frac{r_2}{r_1} - 1}$$



Trapezoid

$$R = \frac{\frac{1}{2}h^2(b_1 + b_2)}{(b_1r_2 - b_2r_1) \ln \frac{r_2}{r_1} - h(b_1 - b_2)}$$

Radius of neutral surface for various cross-sectional shapes

Curved Beam

- Alternative expressions for the normal stress σ_x in a curved beam are

$$\sigma_x = -\frac{My}{Ae(R-y)}$$

and

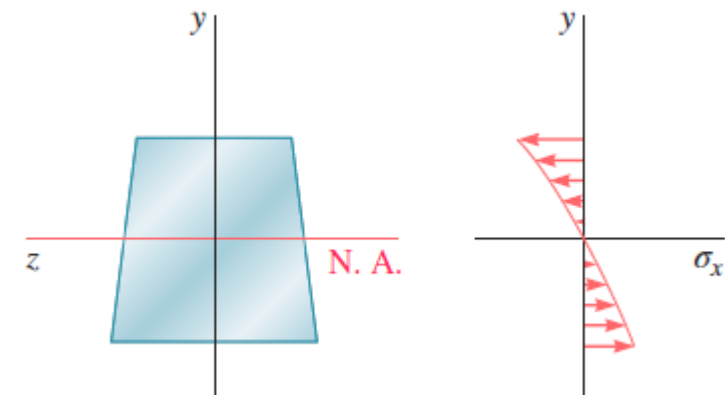
$$\theta' = \theta + \Delta\theta$$

$$\sigma_x = \frac{M(r-R)}{Aer}$$

M: Internal bending moment (positive moment tends to increase curvature).

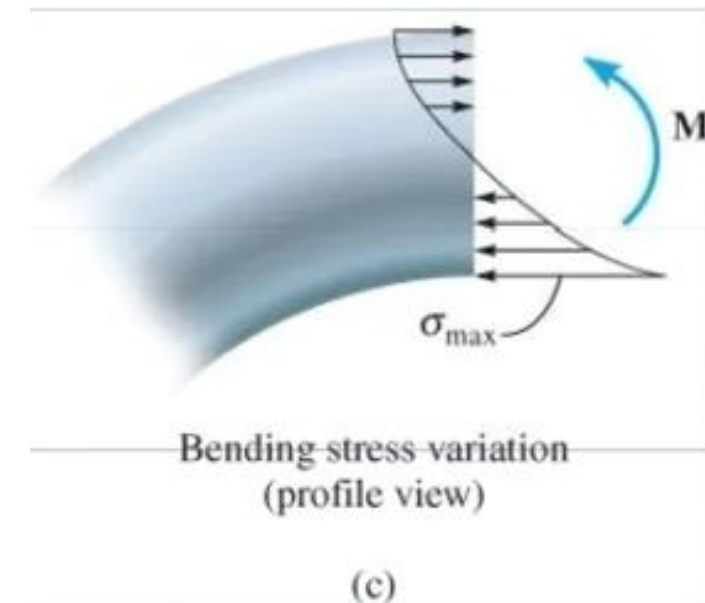
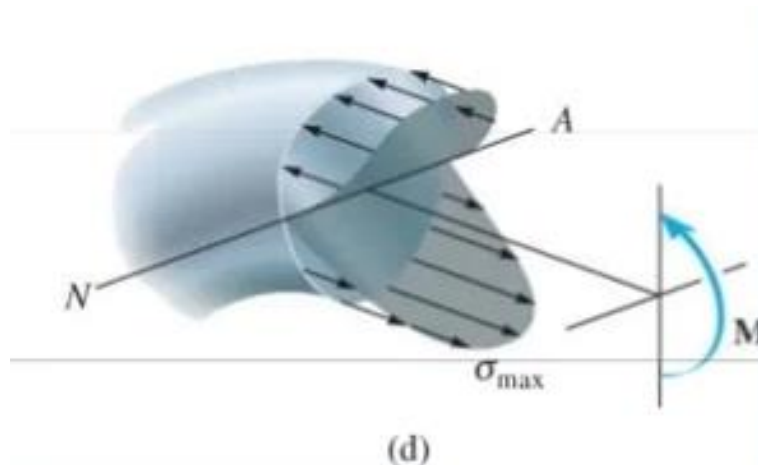
r: Radial distance from C to the point where stress is being calculated.

Relationship shows that, while each transverse section remains plane, the normal strain ϵ_x *does not vary linearly* with the distance y from the neutral surface.



Normal stress in curved beam:

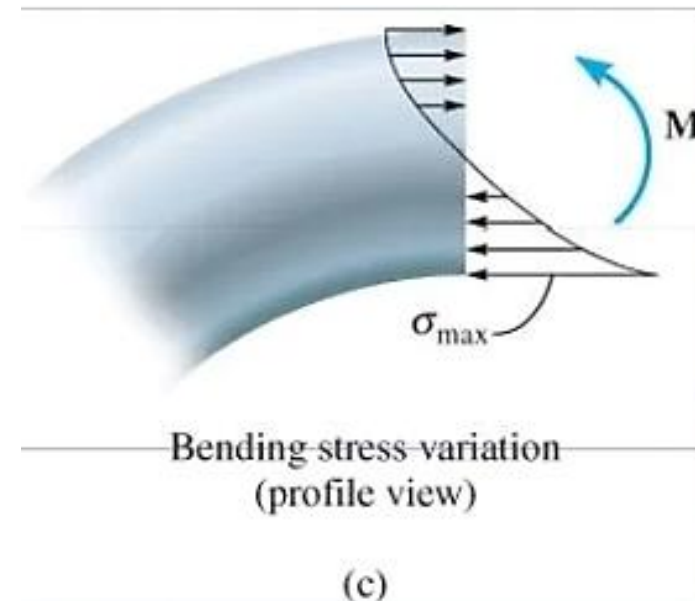
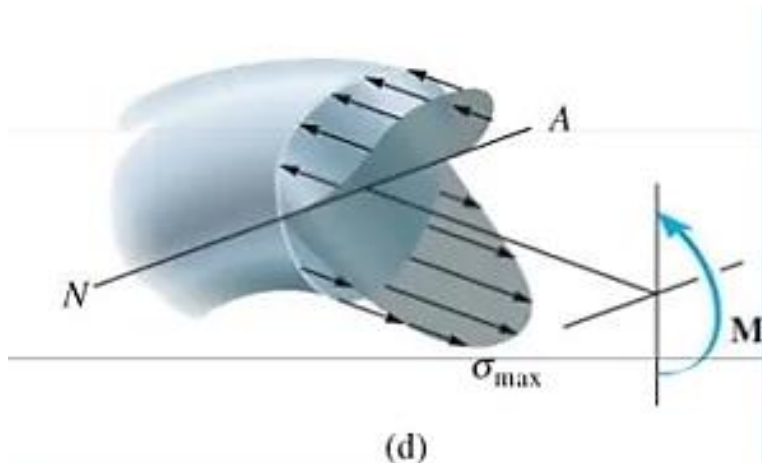
- The stress distribution is as shown , hyperbolic, and is sometimes called circumferential stress
- Radial stress will be created as a result
- If the radius of curvature is greater than 5x the depth of member, flexure formula can be used to determine the stress instead



Curved Beams

IMPORTANT

- Curved beam formula used to determine circumferential stress in a beam when radius of curvature is less than five times the depth of the beam
 - Due to beam curvature, normal strain in beam does not vary linearly with depth as in the case of straight beam. Thus, neutral axis does not pass through centroid of section
-
- Positive stress indicates tensile, negative means



Example

A curved rectangular bar has a mean radius $r = 6$ in. and a cross section of width $b = 2.5$ in. and depth $h = 1.5$ in. Determine the distance e between the centroid and the neutral axis of the cross section.

Solution:

We first derive the expression for the radius R of the neutral surface. Denoting by r_1 and r_2 , respectively, the inner and outer radius of the bar

$$R = \frac{A}{\int_{r_1}^{r_2} \frac{dA}{r}} = \frac{bh}{\int_{r_1}^{r_2} \frac{dr}{r}} = \frac{h}{\ln \frac{r_2}{r_1}}$$

$$R = \frac{h}{\ln \frac{r_2}{r_1}}$$

For the given data,

$$r_1 = \bar{r} - \frac{1}{2}h = 6 - 0.75 = 5.25 \text{ in.}$$

$$r_2 = \bar{r} + \frac{1}{2}h = 6 + 0.75 = 6.75 \text{ in.}$$

Example

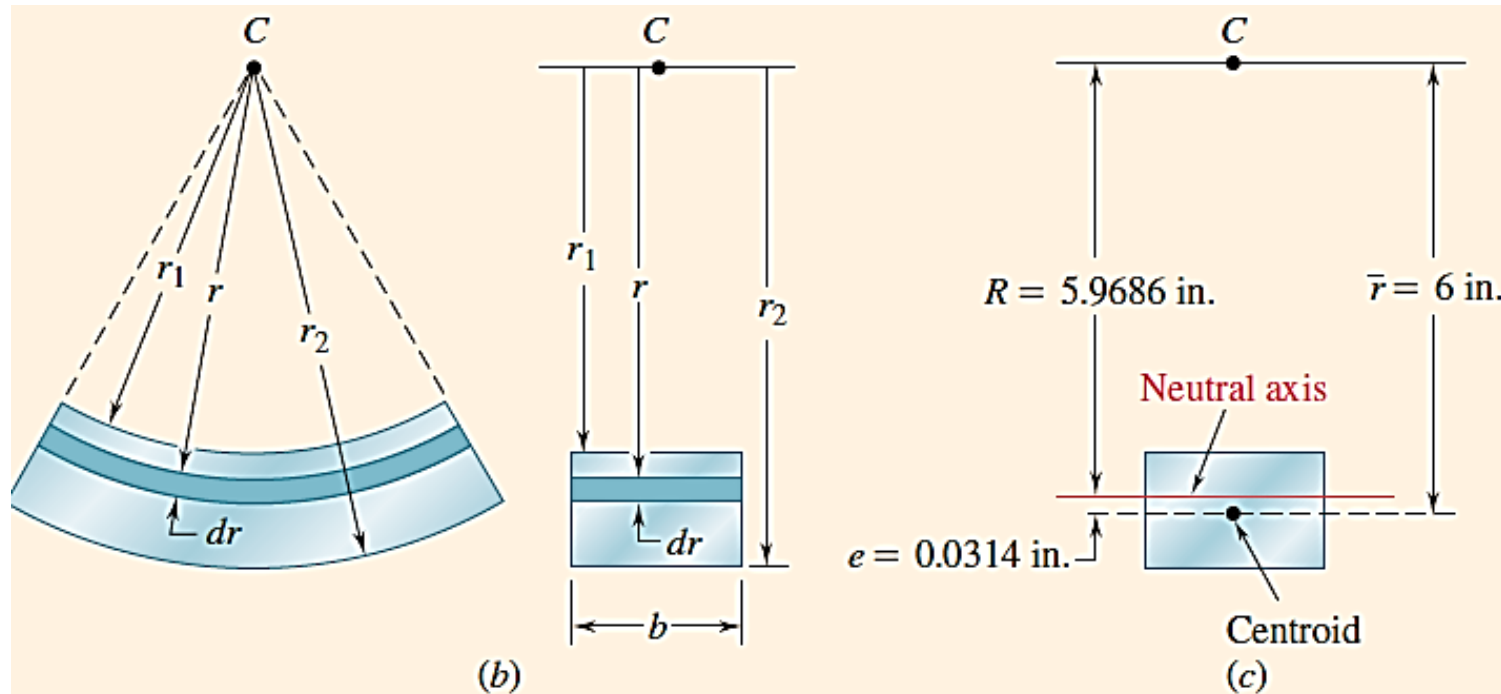
Substituting for h , r_1 , and r_2 into Eq. (4.73),

$$R = \frac{h}{\ln \frac{r_2}{r_1}} = \frac{1.5 \text{ in.}}{\ln \frac{6.75}{5.25}} = 5.9686 \text{ in.}$$

The distance between the centroid and the neutral axis of the cross-section

$$e = \bar{r} - R = 6 - 5.9686 = 0.0314 \text{ in.}$$

Note that it was necessary to calculate R with five significant figures in order to obtain e with the usual degree of accuracy



Example

- Determine the largest tensile and compressive stresses, knowing that the bending moment in the bar is $M = 8 \text{ kip}\cdot\text{in.}$
 - with the given data

$$M = 8 \text{ kip}\cdot\text{in.} \quad A = bh = (2.5 \text{ in.})(1.5 \text{ in.}) = 3.75 \text{ in}^2$$

and the values obtained in for R and e :

$$R = 5.969 \quad e = 0.0314 \text{ in.}$$

First using $r = r_2 = 6.75 \text{ in.}$

$$\begin{aligned}\sigma_{\max} &= \frac{M(r_2 - R)}{Aer_2} \\ &= \frac{(8 \text{ kip}\cdot\text{in.})(6.75 \text{ in.} - 5.969 \text{ in.})}{(3.75 \text{ in}^2)(0.0314 \text{ in.})(6.75 \text{ in.})} \\ &= 7.86 \text{ ksi}\end{aligned}$$

Example

Now using $r = r_1 = 5.25$

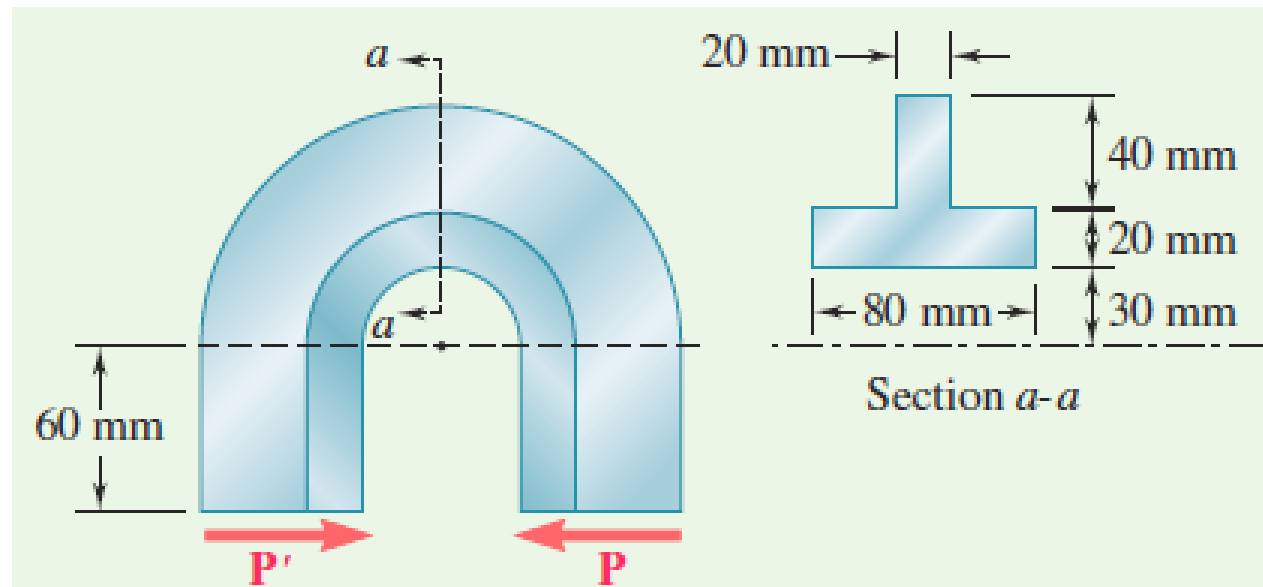
$$\begin{aligned}\sigma_{\min} &= \frac{M(r_1 - R)}{Aer_1} \\ &= \frac{(8 \text{ kip}\cdot\text{in.})(5.25 \text{ in.} - 5.969 \text{ in.})}{(3.75 \text{ in}^2)(0.0314 \text{ in.})(5.25 \text{ in.})} \\ &= -9.30 \text{ ksi}\end{aligned}$$

Remark. Compare the values obtained for $\sigma_{\max}$ and $\sigma_{\min}$ with the result for a straight bar.

$$\begin{aligned}\sigma_{\max, \min} &= \pm \frac{Mc}{I} \\ &= \pm \frac{(8 \text{ kip}\cdot\text{in.})(0.75 \text{ in.})}{\frac{1}{12}(2.5 \text{ in.})(1.5 \text{ in.})^3} = \pm 8.53 \text{ ksi}\end{aligned}$$

Example

- A machine component has a T-shaped cross section and is loaded as shown. Knowing that the allowable compressive stress is 50 MPa, determine the largest force P that can be applied to the component.



STRATEGY:

- The properties are first determined for the singly symmetric cross section. The force and couple at the critical section are used to calculate the maximum compressive stress, which is obtained by superposing the axial stress and the bending.
- This stress is then equated to the allowable compressive stress to determine the force P .

Example

MODELING and ANALYSIS:

Centroid of the Cross Section. Locate the centroid D of the cross section

	$A_i, \text{ mm}^2$	$\bar{r}_i, \text{ mm}$	$\bar{r}_i A_i, \text{ mm}^3$	$\bar{r} \Sigma A_i = \Sigma \bar{r}_i A_i$
1	$(20)(80) = 1600$	40	64×10^3	$\bar{r}(2400) = 120 \times 10^3$
2	$(40)(20) = 800$	70	56×10^3	$\bar{r} = 50 \text{ mm} = 0.050 \text{ m}$
	$\Sigma A_i = 2400$		$\Sigma \bar{r}_i A_i = 120 \times 10^3$	

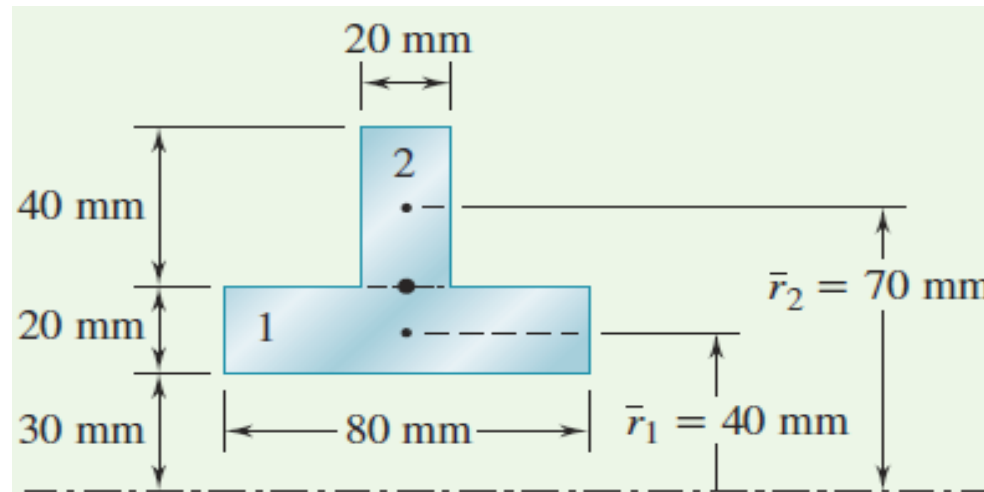


Fig. 1 Composite areas to calculate centroid location.

Example

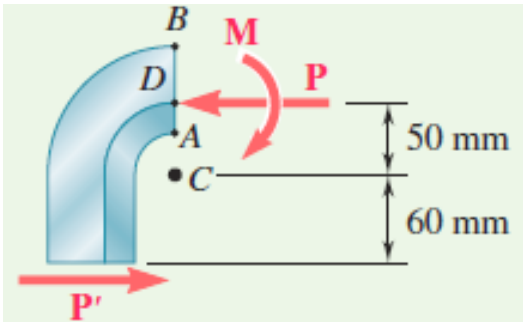


Fig. 2 Free-body diagram of left side.

Force and Couple at D . The internal forces in section $a-a$ are equivalent to a force $\mathbf{P}$ acting at D and a couple $\mathbf{M}$ of moment

$$M = P(50 \text{ mm} + 60 \text{ mm}) = (0.110 \text{ m})P$$

Superposition. The centric force $\mathbf{P}$ causes a uniform compressive stress on section $a-a$. The bending couple $\mathbf{M}$ causes a varying stress distribution

We note that the couple $\mathbf{M}$ tends to increase the curvature of the member and is therefore positive. The total stress at a point of section $a-a$ located at distance r from the center of curvature C is

$$\sigma = -\frac{P}{A} + \frac{M(r - R)}{Aer}$$

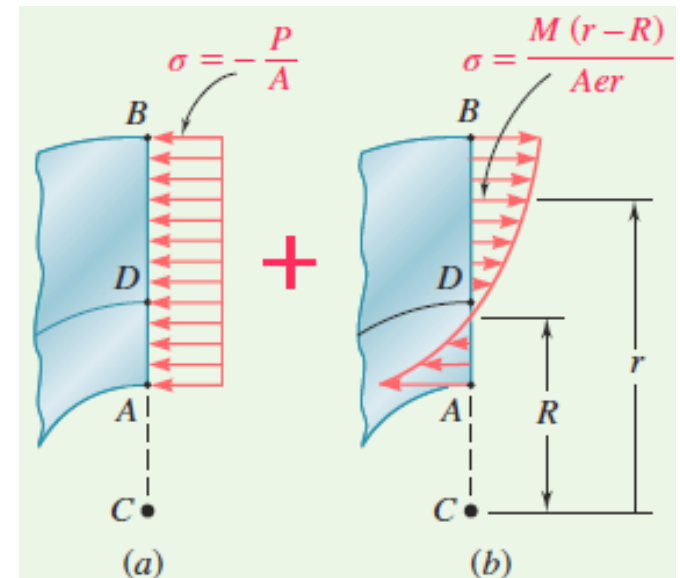
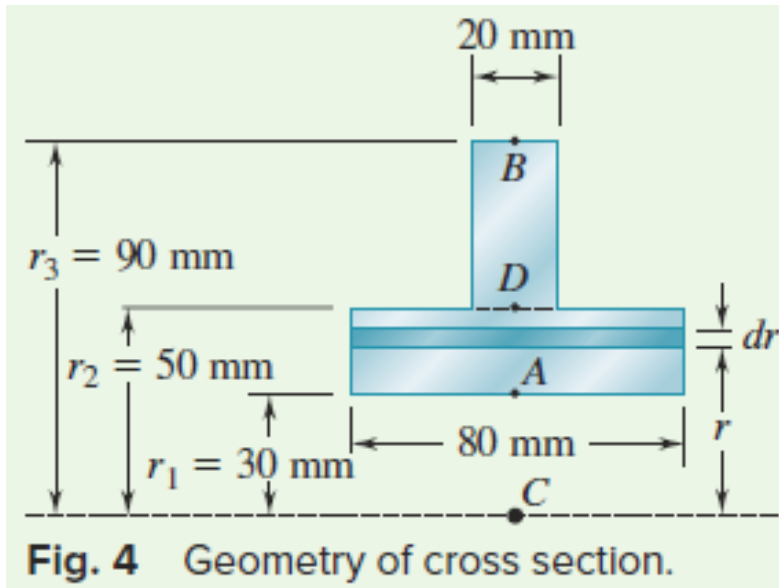


Fig. 3 Stress distribution is the superposition of (a) axial stress and (b) bending stress.

Example



Radius of Neutral Surface. Using Figure, we now determine the radius R of the neutral surface by using

$$R = \frac{A}{\int \frac{dA}{r}} = \frac{2400 \text{ mm}^2}{\int_{r_1}^{r_2} \frac{(80 \text{ mm}) dr}{r} + \int_{r_2}^{r_3} \frac{(20 \text{ mm}) dr}{r}}$$

$$= \frac{2400}{80 \ln \frac{50}{30} + 20 \ln \frac{90}{50}} = \frac{2400}{40.866 + 11.756} = 45.61 \text{ mm}$$

$$= 0.04561 \text{ m}$$

We also compute: $e = \bar{r} - R = 0.05000 \text{ m} - 0.04561 \text{ m} = 0.00439 \text{ m}$

Allowable Load. We observe that the largest compressive stress will occur at point A where $r = 0.030 \text{ m}$. Recalling that $\sigma_{all} = 50 \text{ MPa}$ and using Eq. (1)

$$-50 \times 10^6 \text{ Pa} = -\frac{P}{2.4 \times 10^{-3} \text{ m}^2} + \frac{(0.110 P)(0.030 \text{ m} - 0.04561 \text{ m})}{(2.4 \times 10^{-3} \text{ m}^2)(0.00439 \text{ m})(0.030 \text{ m})}$$

$$-50 \times 10^6 = -417P - 5432P \quad P = 8.55 \text{ kN}$$

Summarize Concept

Concept	Straight Beams ($\sigma = -My/I$)	Curved Beams ($\sigma = M(r-R)/Aer$)
Neutral Axis (N.A.)	Coincides with the centroid of the cross-section.	Shifts toward the center of curvature (away from the centroid).
Bending Stress Distribution	Linear (varies linearly with distance from N.A.).	Hyperbolic (non-linear).
Formula Applicability	Applies when the radius of curvature (R) is very large (nearly infinite).	Must be used when R is small relative to the beam depth (h).